

Modeling and Simulation of the Shape and Interior Pressure of Enveloped Viruses:

Motivated by Orthopoxviruses and Coronaviruses

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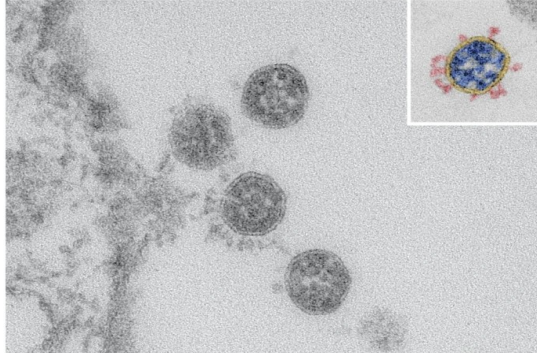


Figure 1 SARS-CoV-2 particles

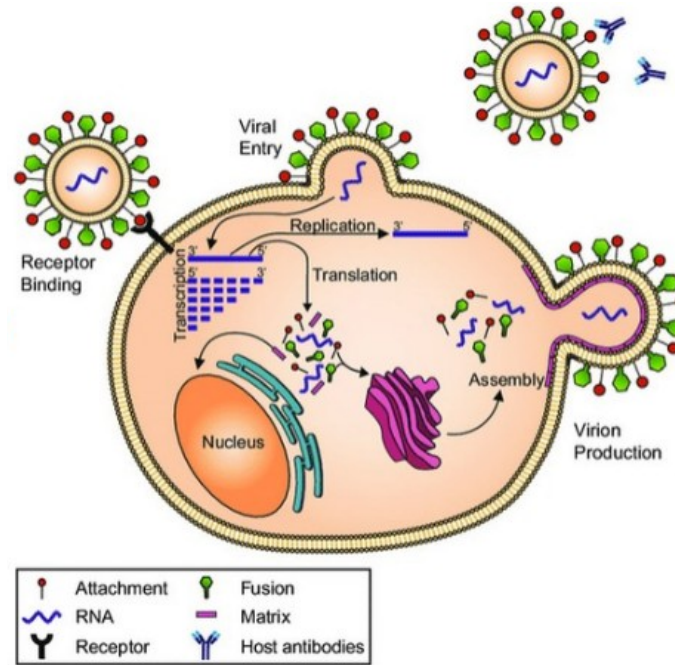


Figure 3 Diagram of the viral life cycle.

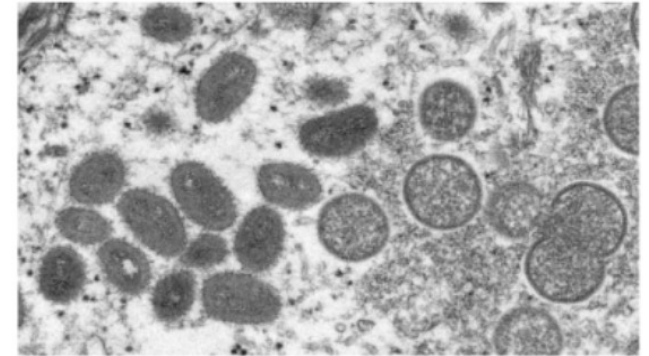


Figure 3 Monkeypox virus particles

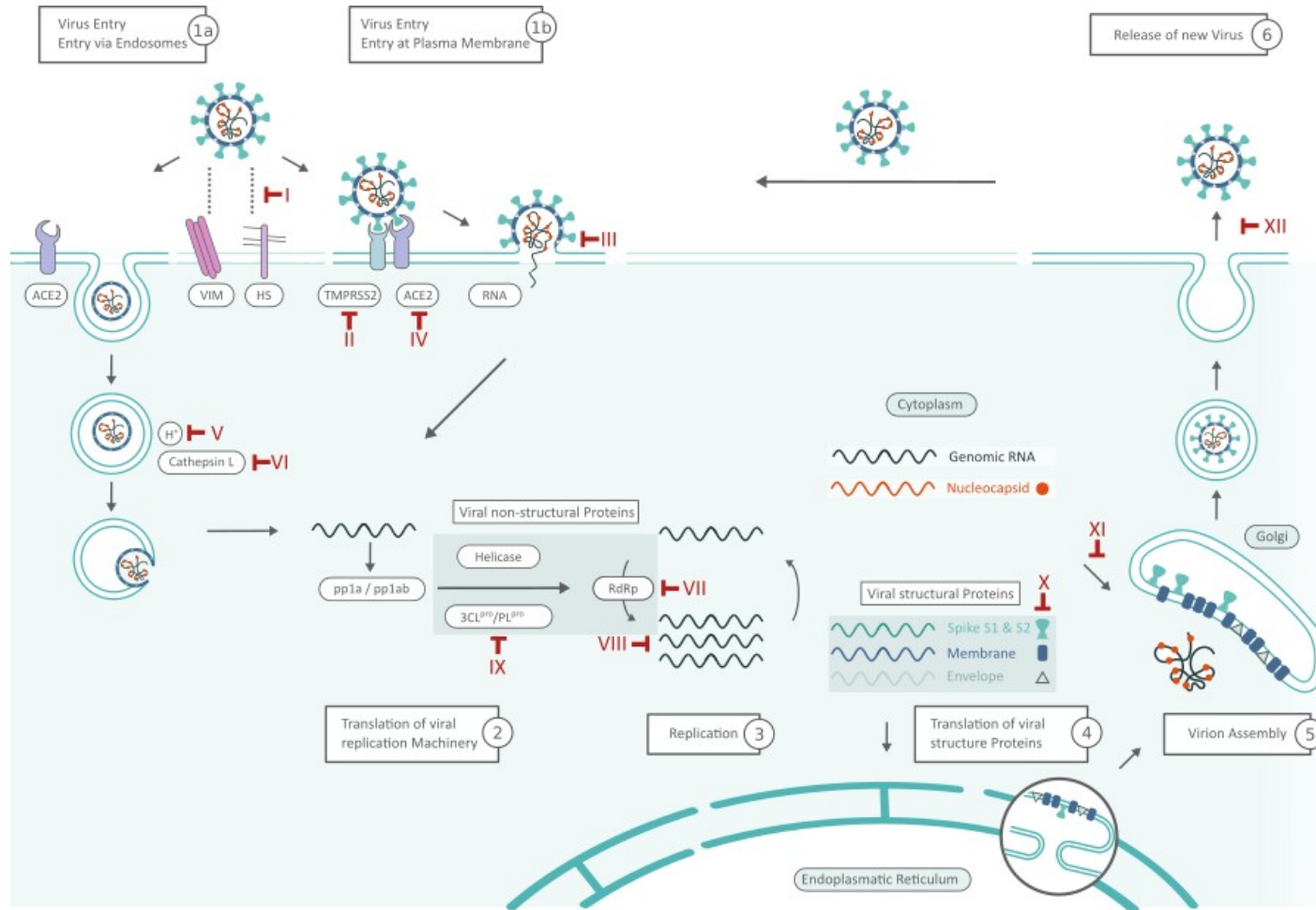


Figure 4 SARS-CoV-2 particles



Figure 5 Coronaviruses. Colorized transmission electron micrograph (TEM) of SARS-CoV-2, the coronavirus that causes COVID-19. The round particles with a fuzzy outer edge show the spike (S) glycoproteins projecting from the viral envelope.

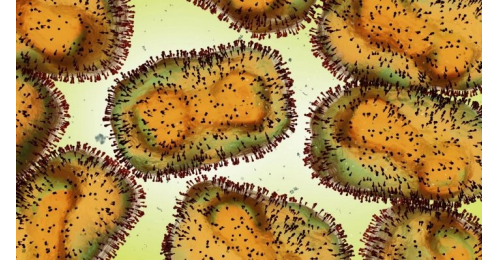


Figure 6 Orthopoxviruses. Colorized transmission electron micrograph highlighting the brick-shaped morphology and surface structures of mature virions.

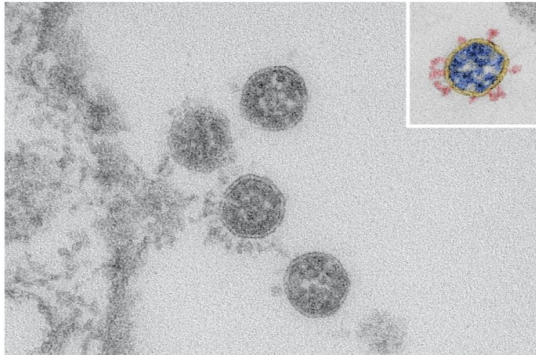


Figure 7 Four SARS-CoV-2 particles imaged by transmission electron microscopy at 135,000x magnification, positioned near the Vero cell surface on the left. The inset displays a color-coded particle with spikes in red, the viral membrane in yellow, and the ribonucleoprotein in blue. Scale bar = 100 nm.[22]

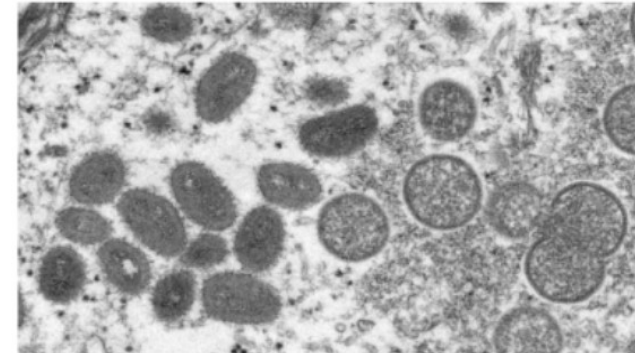


Figure 8. Electron microscopic (EM) image for Monkeypox virus particles. Oval-shaped virus particles are mature, and spherical particles are immature virions.[6]

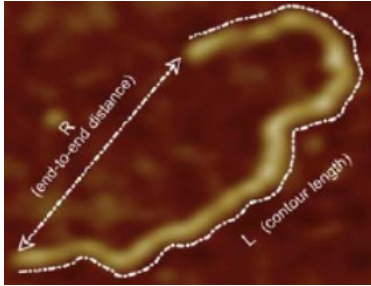


Figure 9 Determination of the contour length and the end-to-end distance for an isolated DNA. [23]

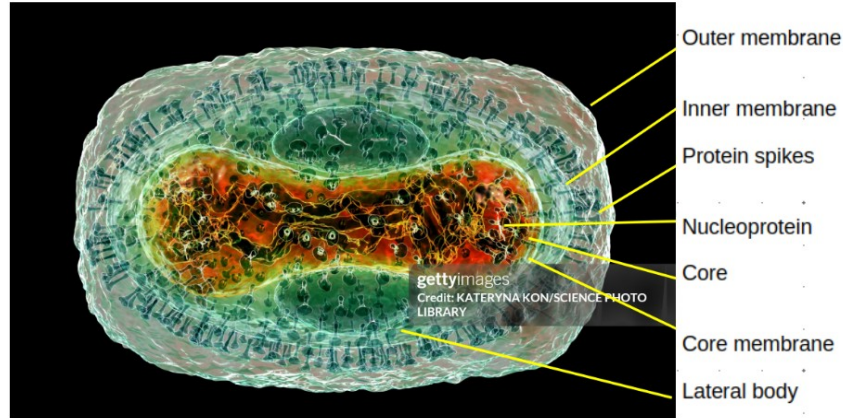


Figure 10 Monkeypox virus particle, Illustration

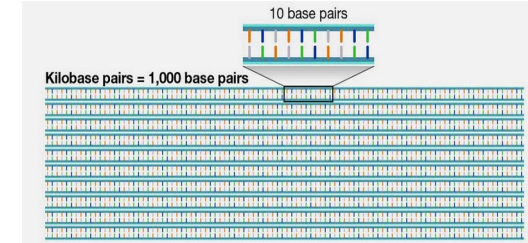


Figure 11. A kilobase (abbreviated kb) is a unit of measurement used to help designate the length of DNA or RNA. One kilobase is equal to 1,000 bases.

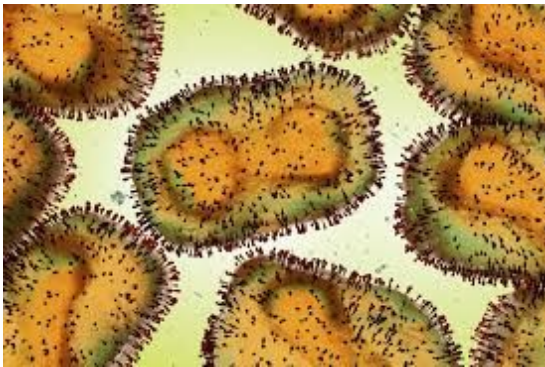


Figure 12 Orthopoxviruses



Figure 13 Coronaviruses

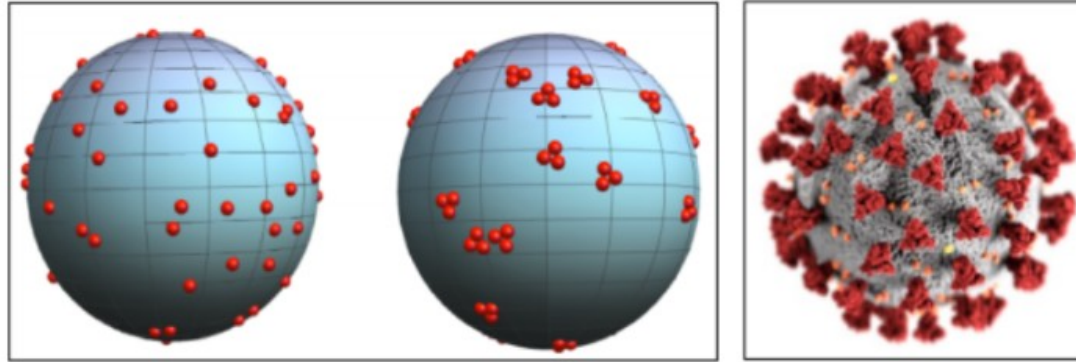


Figure 14 Left image: Simulation of the SARS-CoV-2 spike protein on a spherical surface, illustrating the progression from randomly distributed monomers (Left) to trimer (Right) formations. Right image: SARS-CoV-2 for Covid-19 schematic image (CDC Public Health Image Library)

Coarse-Grained Models (CGM):

It simplifies complex systems by grouping atoms or molecules into larger particles called beads. This allows for large-scale simulations at low computational cost and supports the identification of minimal-parameter models.

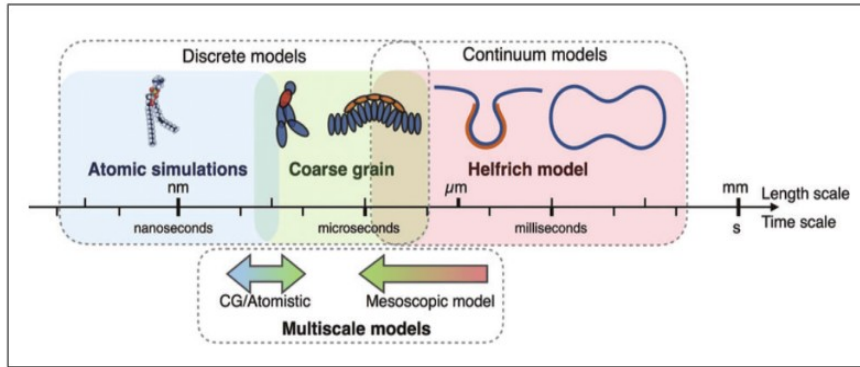


Figure 15 Different computational methods developed to study membrane and genome are valid in different length and time scales.

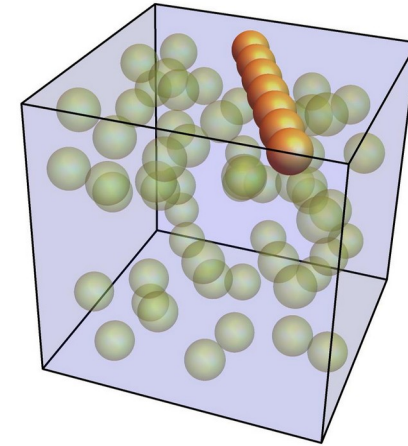


Figure 16 Model of DNA/RNA (Orange Rod) and Organelles (Green Spheres) Confined in a Membrane (Rectangular Prism)

We use a coarse-grained model simulated by the Monte Carlo method. Combine many atoms into one bead to:

- Reduce complexity
- Run much faster simulations
- Study large systems (genomes, membranes, polymers)

Monte Carlo Simulations: Using random sampling to solve math and physics problems

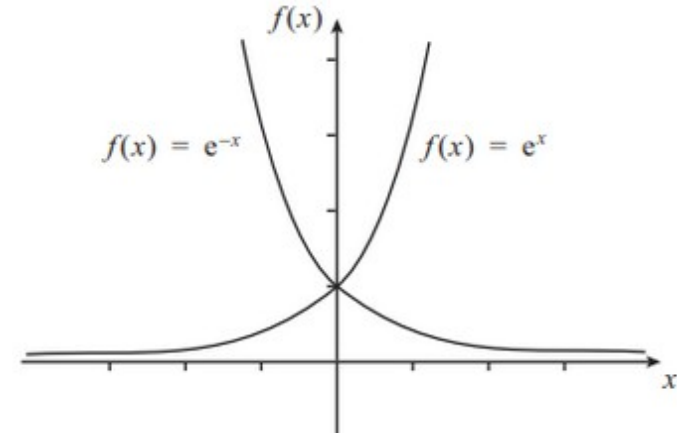
The Metropolis acceptance probability P is defined as:

$$P = \begin{cases} 1 & \text{if } \Delta E \leq 0 \\ e^{-\Delta E/(k_B T)} & \text{if } \Delta E > 0 \end{cases}$$

Metropolis rule:

$$P = \min(1, e^{-\beta \Delta E})$$

- If $\Delta E < 0$, accept the move.
- If $\Delta E > 0$, generate a random number $r \in (0, 1)$
 - If $r < P$, accept the move.
 - If $r > P$, reject the move.



To ensure that any two particles i and j do not overlap, we compute the squared center-to-center distance

$$d_{ij}^2 = (x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2. \quad (1)$$

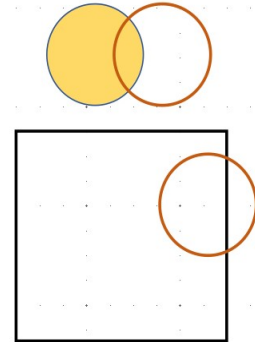
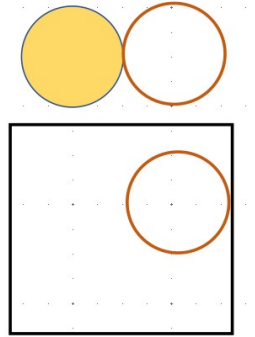
For the particles with the diameter σ , overlap is rejected by imposing the condition

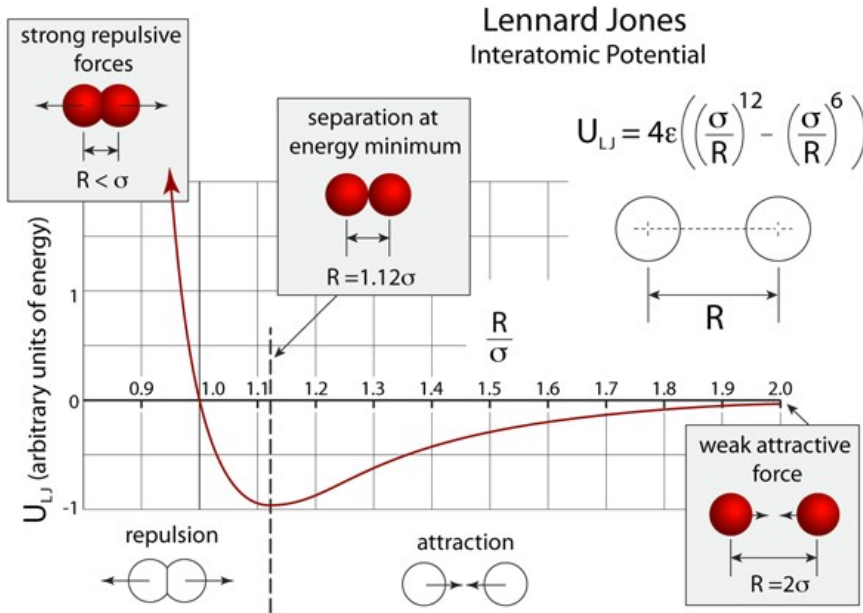
$$d_{ij} < \sigma \quad (2)$$

$$-\frac{L}{2} + r \leq x \leq \frac{L}{2} - r, \quad (3)$$

$$-\frac{L}{2} + r \leq y \leq \frac{L}{2} - r, \quad (4)$$

$$-\frac{L}{2} + r \leq z \leq \frac{L}{2} - r. \quad (5)$$





Attraction between genome and intracellular components

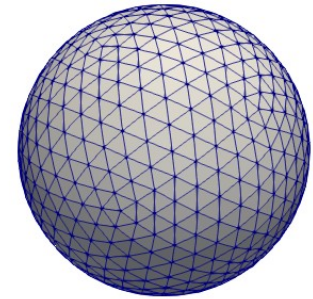
$$U_{LJ}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right], \quad r \geq 2^{1/6}\sigma$$

σ is the distance at which the Lennard–Jones potential crosses zero, $U_{LJ}(r = \sigma) = 0$.

(1) Continuum description of membrane bending energy (theoretical):

$$F_{\text{full mem}} = \int_S \left[\underbrace{\frac{\kappa}{2} (2H - C_0(\vec{r}))^2}_{\text{(1) spontaneous-curvature-modified bending}} + \underbrace{\bar{\kappa} K}_{\text{(2) Gaussian-curvature term}} \right] dA$$

$$+ \underbrace{\lambda A}_{\text{(3) area constraint}} + \underbrace{pV}_{\text{(4) volume constraint}} .$$



(2) Discrete angle-based formulation for simulations (practical):

$$E_{\text{bend}} = \kappa \sum_{\langle i,j \rangle} (1 - \cos \theta_{ij}) + \lambda A + pV$$

where θ_{ij} is the dihedral angle between neighboring triangles

(1) Continuum description of membrane bending energy (theoretical):

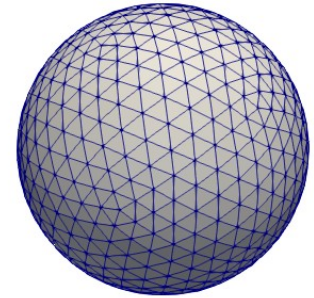
$$F_{\text{full mem}} = \int_S \left[\underbrace{\frac{\kappa}{2} (2H - C_0(\vec{r}))^2}_{(1) \text{ spontaneous-curvature-modified bending}} + \underbrace{\bar{\kappa} K}_{(2) \text{ Gaussian-curvature term}} \right] dA$$

$$+ \underbrace{\lambda A}_{(3) \text{ area constraint}} + \underbrace{pV}_{(4) \text{ volume constraint}} .$$

(2) Discrete angle-based formulation for simulations (practical):

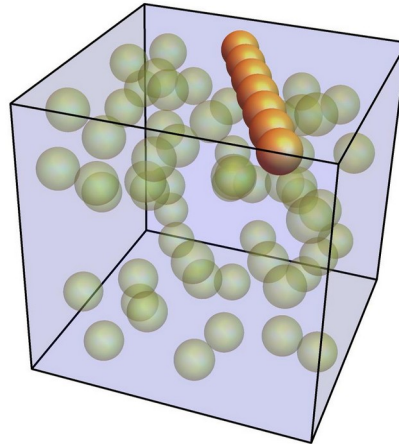
$$E_{\text{bend}} = \kappa \sum_{\langle i,j \rangle} (1 - \cos \theta_{ij}) + \lambda A + pV$$

where θ_{ij} is the dihedral angle between neighboring triangles



A simple liquid model describes a many-particle system by adding together simple pairwise interaction potentials between particles. The total energy is the sum of the membrane bending energy and basic pairwise interactions.

$$E = E_{\text{mem-bend}} + \sum_{\text{DNA-bonds}} U_{\text{LJ}} + \sum_{\text{DNA-mem}} U_{\text{LJ}} + \sum_{\text{DNA-crowder}} U_{\text{LJ}} + \sum_{\text{crowder-crowder}} U_{\text{LJ}}$$



NPT ensemble (constant N,P,T)

$$P_{\text{accept}} = \min \left(1, \exp \left[-\frac{\Delta E + p\Delta V - Nk_B T \ln \left(\frac{V_{\text{new}}}{V_{\text{old}}} \right)}{k_B T} \right] \right)$$

- $\Delta E = E_{\text{new}} - E_{\text{old}}$: change in system energy due to a trial move
- k_B : Boltzmann constant
- T : temperature

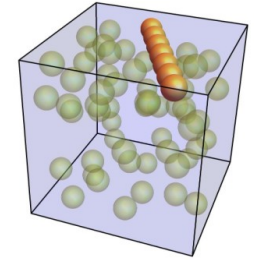
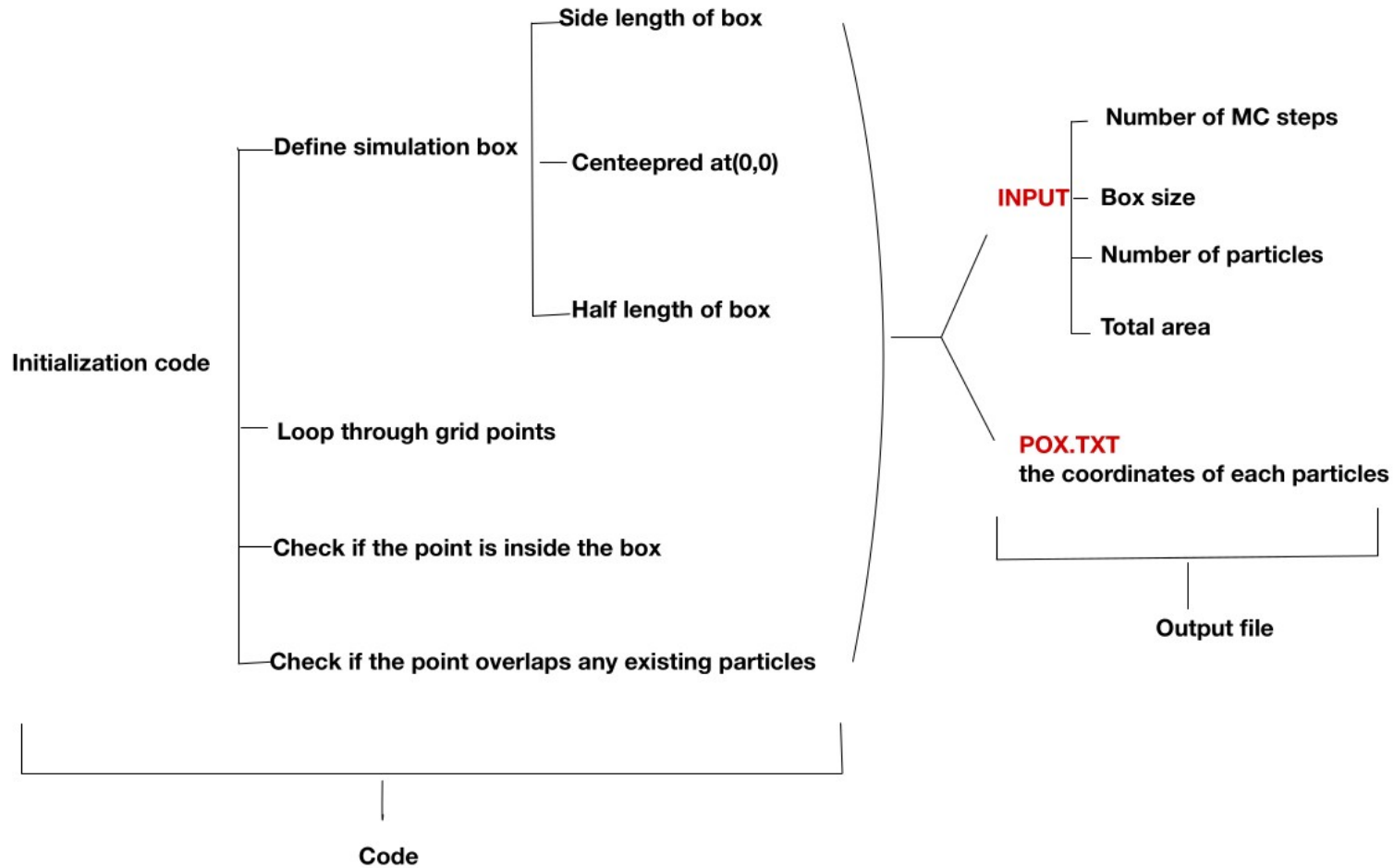
NVT ensemble (constant N,V,T)

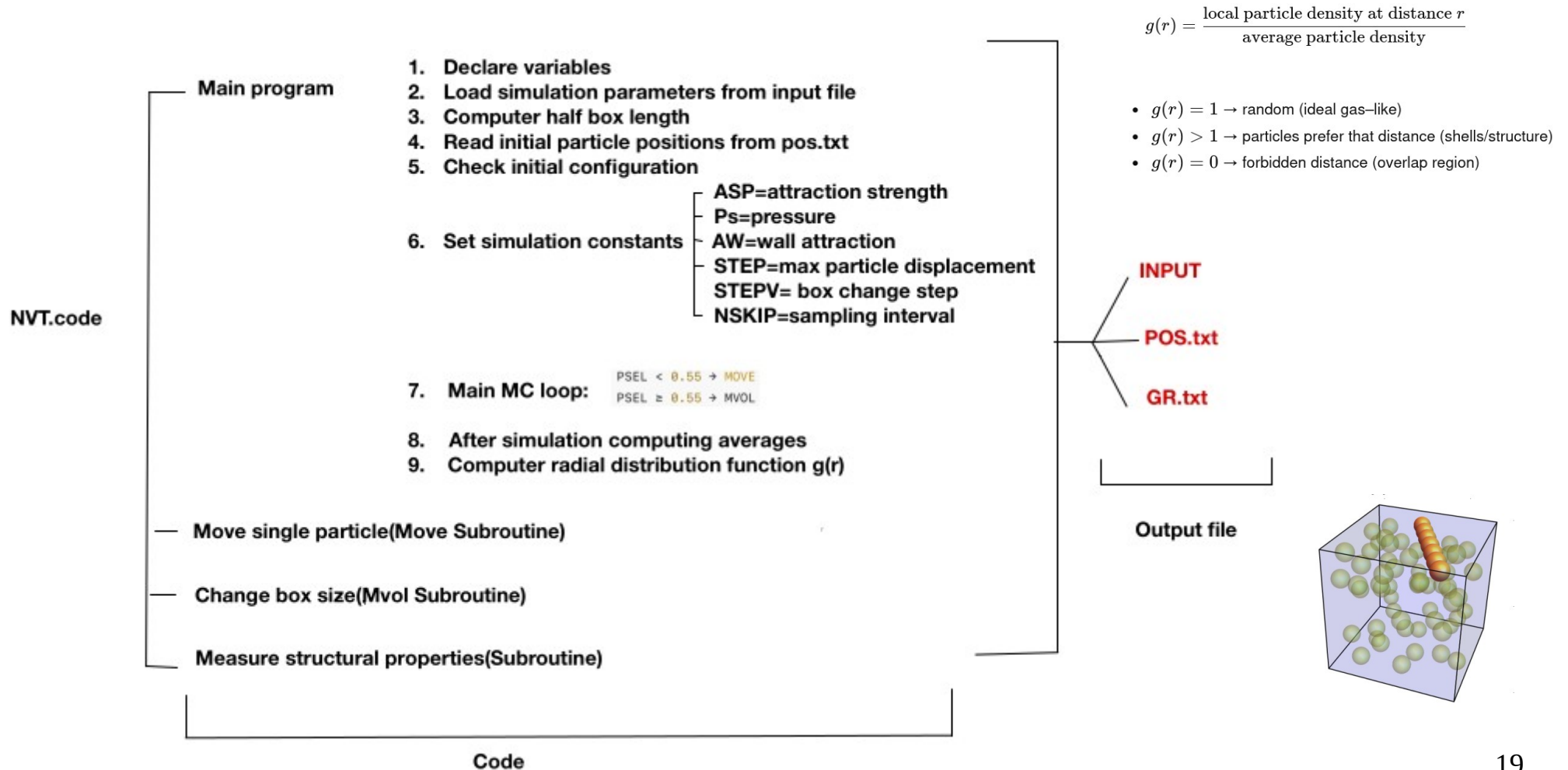
$$P_{\text{accept}} = \min \left(1, \exp \left[-\frac{\Delta E}{k_B T} \right] \right)$$

- $p \Delta V$: mechanical work associated with changing the system volume
- $\Delta V = V_{\text{new}} - V_{\text{old}}$
- The logarithmic term arises from the change in configurational entropy due to volume scaling of particle coordinates
- N : number of particles

Hypothesis 1: Virus shape is determined by genome packing and by interactions between the genome, internal components, and the viral membrane.

Hypothesis 2: During maturation, increased genome condensation raises internal pressure, which drives a transition from spherical to elongated virus shape.





NPT.code

Main program

Program header
Seed random number generator
Read INPUT file
Computer half box sizes
Compute box volume for packing fracture
Read particle positions
Check all particles inside the box
Check for overlap

Set simulation parameters

```

PS = 3.D0      ! Pressure
ASP = 4.D0     ! Attraction strength
AW = 0.D0      ! Wall attraction strength
STEP = 0.00035*(ALX+ALY+ALZ)/3
STEPV = 0.00007*(ALX+ALY+ALZ)/3
    
```

Main MC loop
Sample every 50 steps
After simulation print statistics
Write update INPUT and POS.TXT
Computer g(r)

$$E_{\text{wall}} \propto -\left(\frac{1}{(x - HALX - 0.5)^6} + \dots\right) \text{ accept if: } rand < \exp(E_{\text{old}} - E_{\text{new}})$$

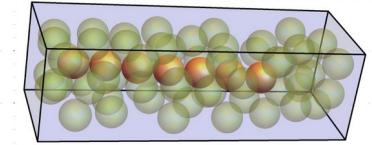
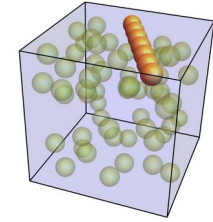
Move subroutine

Mvol subroutine

Create histogram for g(r)

$$r_{ij}^2 = (x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2$$

Reject move if $r_{ij}^2 + 10^{-6} < 1$



$$\Delta H = P(V' - V) - N \ln \left(\frac{V'}{V} \right) + (E' - E)$$

55% of the time → attempt a particle move

45% of the time → attempt a volume move

Too many volume moves makes the simulation unstable.

Too few volume moves and the pressure cannot equilibrate.

A common rule is 50-90% particle moves, the rest volume moves.

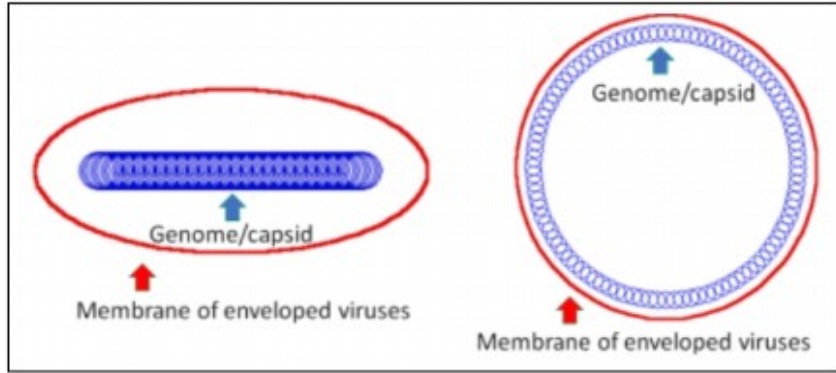


Figure 17 Preliminary 2D model studies to investigate the effect of the geometry of a genome on the shape of a virus. A rod-like genome induces an elliptic shape whereas a circular genome leads to a circular shape

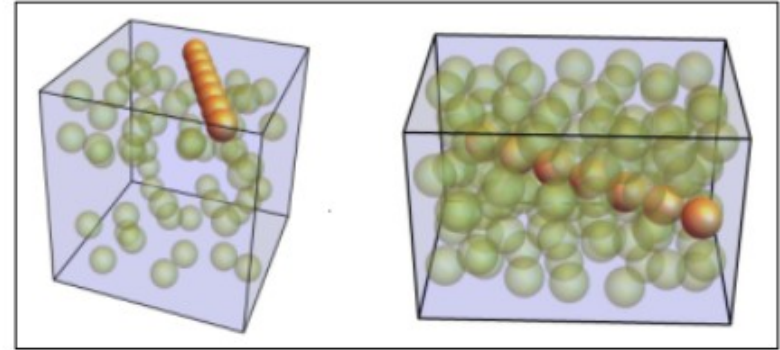


Figure 18 Initial cubic configuration (left) and final elongated cuboid configuration (right) obtained after sequential NVT equilibration followed by NPT simulation, consistent with the morphology of Monkeypox (Orthopoxvirus) particles.



A super-ellipsoid is a smooth 3D shape defined by the equation below

$$\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n + \left(\frac{z}{c}\right)^n = 1$$

The equation generalizes simple shapes by changing parameter n .

- a, b, c : Control the size along the three axes → long, flat, or brick-like shapes
- n (shape parameter): Controls how rounded or box-like the surface

Table 1 Effect of the shape parameter n on superellipsoid geometry

n value	Shape behavior
$n = 2$	Ellipsoid (sphere if $a = b = c$)
$n > 2$	Rounded rectangular or brick-like shape
$n \rightarrow \infty$	Approaches a rectangular prism
$n < 2$	More pointed, diamond-like shape

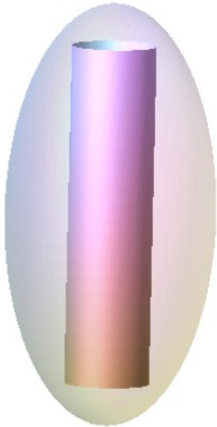


Table 2: Comparison of Rectangular Prism and Superellipsoid Shape Models

Feature	Rectangular Prism Model	Superellipsoid Model
Realism	Sharp edges, less realistic	Smooth edges similar to real orthopox viruses
Curvature	No curvature; bending energy cannot be computed	Curvature varies spatially
Surface Area	Easy to compute analytically	Requires numerical integration
Shape Flexibility	Only cuboid shapes possible	Can smoothly interpolate between sphere, ellipsoid, and rounded rectangular shapes
Shape Transitions	It can only study cube to rectangle transition	Allows gradual and continuous shape transitions
Cost	Low computational cost	Higher computational cost
Overall Suitability	Less realistic virus modeling	More realistic and flexible for virus shape modeling

The Superellipsoid model may help explain:

Why

COVID-19 virus particle stays more spherical

Monkeypox virus particle becomes dumbbell-shaped

How:

Crowding + container curvature reshape genome packing

Why:

Immature viruses are round

Mature orthopoxviruses are brick-like



Figure 17 Coronaviruses.

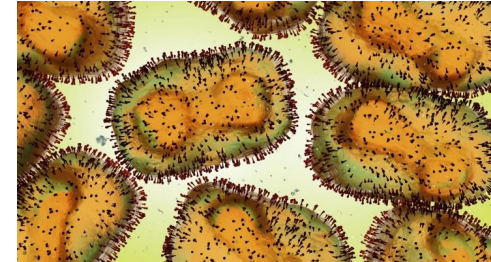


Figure 18 Orthopoxviruses.

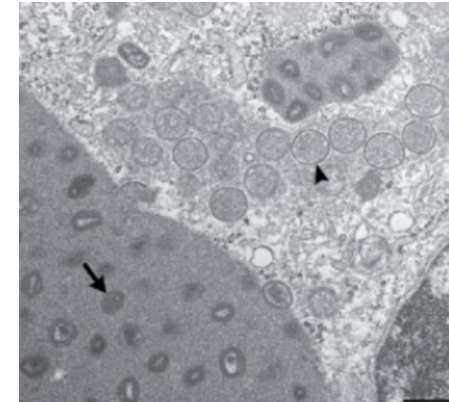


Figure 19 The upper box in Panel A at a higher magnification. Type A viral inclusion bodies, which are observed in infections caused by some orthopoxvirus species, are composed of proteinaceous material and mature virus particles (arrow). Type B viral inclusion bodies, also known as viral factories, are composed of spherical, immature virus particles (arrowhead).

Goal 1: Build a Simple Virus Model

Build a simple model of a genome inside a soft membrane.

- Model the genome as connected beads.
- Model the membrane as a flexible surface.
- Include basic weak attraction.
- Study how the genome pushes on and deforms the membrane.

Goal 2: Study What Controls Virus Shape

Use the model to test what physical factors change virus shape.

- Change genome length (short vs long).
- Change genome stiffness (soft vs stiff).
- Change membrane size and flexibility.
- Measure changes in shape, pressure, and genome position.

Goal 3: Explain Shape Changes in Real Viruses

Use the model to explain experimental observations.

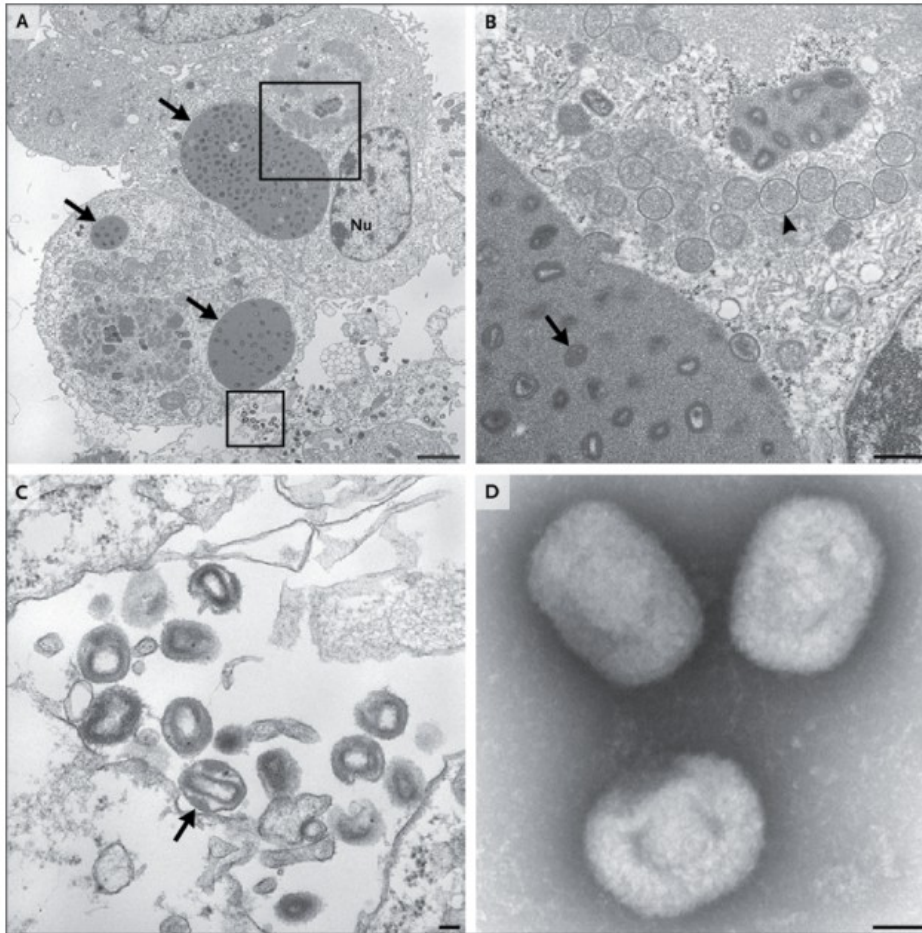
- Explain why COVID-19 particles stay mostly spherical.
- Explain why Monkeypox particles are elongated.
- Explain the change from spherical immature particles to elongated mature particles.

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Thank you!



The Novel Orthopoxvirus on Transmission Electron Microscopy. Thin-section electron microscopy and negative-stain electron microscopy were performed, as previously described . 20,21 Panel A shows type A viral inclusion bodies (arrows) in infected cell-culture cells on thin-section electron microscopy. Nu denotes the cell nucleus. The scale bar represents 2 μ m. Panel B shows the area in the upper box in Panel A at a higher magnification. Type A viral inclusion bodies, which are observed in infections caused by some orthopoxvirus species, are composed of proteinaceous material and mature virus particles (arrow). Type B viral inclusion bodies, also known as viral factories, are composed of spherical, immature virus particles (arrowhead). The scale bar represents 500 nm. Panel C shows a higher magnification of the extracellular mature virions shown in the area in the lower box in Panel A. A lateral body within a maturing virus particle (arrow) is shown. The scale bar represents 100 nm. Panel D shows brick-shaped virions, with short, whirled filaments on the surface on negativestain electron microscopy. The scale bar represents 100 nm.

Vora, Neil & Li, Yu & Geleishvili, Marika & Emerson, Ginny & Khmaladze, Ekaterine & Maghlakelidze, Giorgi & Navdarashvili, Archil & Zakhashvili, Khatuna & Kokhraidze, Maka & Endeladze, Marina & Mokverashvili, Gela & Satheshkumar, P.S. & Gallardo-Romero, Nadia & Goldsmith, Cynthia & Metcalfe, Maureen & Damon, Inger & Maes, Edmond & Reynolds, Mary & Morgan, Juliette & Carroll, Ph.D.. (2015). Human Infection with a Zoonotic Orthopoxvirus in the Country of Georgia. *New England Journal of Medicine*. 372. 10.1056/NEJMoa1407647.

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